



DIGITAL TRANSFORMATION

UNDERSTANDING BUSINESS GOALS, RISKS, PROCESSES, AND DECISIONS

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1. Introduction

The starting point for this book is that digitisation works as a catalyst for society and the organisations therein. A catalyst is something that increases the speed of a process without itself being consumed. It simply produces results faster. Digitisation, if used in the wrong way, can also make a bad situation worse. And it can make what functions well function even better. It is therefore absolutely crucial to start by making clear—or finding out—what one wishes to accomplish, and how well the way that the organisation is planned suits the realisation of that goal. Starting to run will not help an individual headed in the wrong direction. Quite the contrary. But as both the surroundings and the internal operations increasingly build on digitisation, it is important that those who work within an organisation can identify how digitisation affects its operations—and based on that, which technical solutions (IT) are needed to capture the possibilities offered by digitisation, and to avoid its pitfalls.

This book thus deals with value, organising, and digitisation—how to direct and organise one's business so as to take advantage of the possibilities that transformative use of IT, digital transformation, can offer, and not just temporarily and in the short run, but sustainably and in the long run (strategically). By digitisation, we mean the use of modern IT to create, deliver and use products (goods, services, and combinations of the two). When we speak of benefit and value, we do not simply mean asset growth for shareholders (shareholder value), but also value for other stakeholders—co-workers, customers, suppliers, and the organisational surrounding. Sometimes a venture can be positive to many, and sometimes interests clash. We strive to note discrepancies, but our main perspective is from within the organisation looking out; how can those in an organisation work to strategically further the business results? We focus on the organisation and the business it conducts; we

do not specifically look at individuals. We write about strategy and targeting goals, but this is not a book about the business leader or the brilliant technician as an omniscient hero. Nor are we primarily external observers, studying and assessing how the organisation affects others. But when adopting a perspective from within, we do so well aware that all organisations act in a context. Some parts of this context are in direct interaction with the organisation; others indirectly influence or are influenced by it. And yet others are so far from it that there is no discernible connection.

Given the focus on digitisation, our interest is directed at the digital and at digitisation as part of conducting business, not as technology in itself, or with a primarily technical focus. No doubt that machine, system, and information architectures all play a role in what can be accomplished, but we do not delve into the technical issues. And “IT” (denoting any technology for information transfer) has always been around in society: cuneiform writing, smoke signals, printing presses, abacuses, telegraphs, radio, computerised data handling, telecommunication, GPS, etc. Today, IT, or sometimes “modern IT”, if one wishes to be more precise, is primarily used to signify that which is connected with computers: the hardware, but also the software and what the computers accomplish when the programs are run—computers as tools. Your smartphone and what it can assist you in doing is “modern IT”. So, too, is much of the control of your modern cooker, and the card reader on the bus.

Our book is about navigating and handling a world with an abundance of data—and sometimes also a lack of relevant data. It is also about navigating a world where products increasingly build on digitisation. Music reaches us as files—resident or streamed—and digital music players (often as a part of a smartphone); the functions of a car are increasingly based on software and digital sensor data; money is increasingly digital, even in transactions between individuals; the airplane engine, with advanced software controls, is rented per flight hour (which are recorded by software); the forklift truck registers how it is driven, to provide a basis for upkeep, but also to identify needs for driver training; socialising and collaboration are increasingly digitised—via social network media, collaboration software, telephone, and video calls, and via the ability to digitally access the material we discuss.

Broadly speaking, digital transformation is the integration of digital technology into all areas of societal activities and processes. Digital transformation is about transforming organisational and societal structures by introducing digital information systems and flows as well as managing such structures and routines utilising these flows. Thus, digital transformations rest heavily on digital information systems, which have long been used to support various activities in organisations and have steadily reduced processing and communication times, but are now changing virtually all sectors of societal activities. Organisations can be geographically dispersed, and agents of various kinds can still communicate, collaborate, and coordinate their activities through standardised and automated dedicated digital platforms. Computation, storage, and transmission capacity are standardised services that can be bought from a host of global suppliers. The possibilities for collecting, analysing, and reporting data using a large set of tools and techniques have increased dramatically with the emergence of business intelligence tools, competitive intelligence, dashboards, data warehousing, data mining, and big data technologies, which have emerged and changed societal activities at large.

Digital transformation has changed product and service delivery in a fundamental way by changing the technological and cultural environment. It is, in principle, affecting everything since both the surroundings and the internal operations in many activities are increasingly built on digital solutions. As a consequence, digital transformations have fundamentally changed, and continue to change, our lives. Naturally, there are a lot of accompanying challenges on different levels. The surveillance capacities are increasing, privacy issues become alarming, social inequalities are emphasised, and although globalisation offers many opportunities, it also contains some severe issues for people that perceive that they are losing understanding, overview, and control. It is therefore crucial for decision-makers in societies and organisations as well as for citizens at large to regain control of the development and its substantial factors to avoid the otherwise inevitable alienation effects. It is therefore necessary to obtain the means to be able to identify how digital transformation affects societal operations in various contexts and to take command in a sustainable way of the challenges and possibilities it offers so as

to navigate, control, and manage effectively in our world with its data abundance and control systems.

During this transformation, societies, organisations, and individuals would be wise to define their goals and in which manner digital solutions should be utilised to achieve them. Likewise, they should strive to find means to control their internal and external environments in increasingly complex contexts. Here, processes for adequate decision-making are of great importance. Otherwise, it will become costly in a variety of aspects: financial, integrity, efficiency, decision-making power, etc. Despite access to large amounts of information, we still have limited capacity to understand what we need and how to achieve our goals. Consequently, we need support for analysing our data and formulating our strategies in order to take control of the development. Nevertheless, the situation today is that most organisations do not have any structured procedures at all for handling decisions. Both methods and the knowledge by which to identify and analyse even fairly simple problems are lacking. Whatever people tend to believe, making adequate decisions is difficult but can be substantially facilitated by a methodical approach since human judgement, in many cases, is simply insufficient. Decision competence must be enhanced in several respects.

The possibilities for digital communication also decrease the need to build organisations according to geographical proximity; it is possible to communicate, direct, and monitor at a distance. In turn, this makes it possible for a particular organisation to be more geographically dispersed and facilitates collaboration between specialised organisations. Production, payroll, recruiting, product development, customer support and helpdesk, sales: all of these functions were previously natural parts of an organisation, and can now be specialties of focused organisations that, by buying and selling services, can collaborate as a coordinated unit, a virtual organisation delivering products to external customers.

What, then, is an organisation? A classical answer is that it is a cooperation to achieve a goal. People organise in order to achieve together what they cannot achieve on their own. One purpose can be simply socialising and human contact. But here, we concentrate on businesses that produce goods and services for customers, users, or members, who in turn can be individuals or organisations. In a juridical sense, an organisation can of course be owned and run by a single

individual, a self-employed person. Most organisations conform to this type. But we direct our attention here to the larger ones, those composed of more than one person. Not all organisations are for-profit, and we will mix examples from for-profit enterprises with examples from the public and non-profit sectors. (We use the word ‘organisation’ in reference to all types of businesses. When we specifically mean for-profit businesses, we use the word ‘company’.) When we talk of organising, we mean the creation, entertaining, and changing of organised ventures—which may be a part of an organisation, an entire organisation, or a collaboration between organisations. (Organising is the subject of Chapter 4.)

What is the digitisation intended to support? A strategic view of digitisation needs to start by addressing this issue.

1.1. Roles of Digitisation in Operations

Product development, production, project management and coordination, supplier communication, and customer communication: which functions in the business should really benefit from IT, and how? When digitisation more clearly became an important business issue, people started to think about which digitisation steps were important (and how), from capture to distribution of data, to the production-chain steps from orders and inbound logistics, to sales and outbound logistics. The point of departure is a physical production organisation. What can be digitised? That which is already clear data in paper (or similar) format does not require much imagination. In 1967, the Swedish public-sector telephone operator (Televerket) proudly announced to its customers that “computers will be introduced for handling telephone bills”. But when many organisations have realised how to efficiently digitise the handling of such data, it will not provide any strategic benefits. However, it can be a disadvantage not to keep up with such development, if it provides clear benefits. Just imagine a bank not offering Internet banking to its customers.

The control of machines and of processes are obvious targets for digitisation, in turn providing possibilities for automation. Numerically controlled lathes, paper machines and increasing numbers of robots are now commonplace. Self-driving vehicles are increasingly being introduced into the consumer and professional markets and are expected to affect how we travel and ship goods.

After data and control, one next step was to think about what could be digitised in a good or a service. Watches, copying machines, and telephones are examples of physical goods that are now digitised. Analogue models have become curiosities. In services, delivery (of letters, music, games, and film) has undergone considerable change. Physical letters have been replaced by electronic messages. We want instantaneous, online access to newspapers, film, and music. Nowadays, we expect to be able to track the progress of a physical delivery or journey digitally. And in manufacturing, industry 4.0 is a concept for supply chains and production equipment that is connected and coordinated through standardised and automated digital communication. The transition from physical or analogue to digital can be expected to continue at the pace at which new possibilities for increased efficiency can be found—and sometimes at the pace at which new possibilities can be found, even if they do not lead to clearly increased efficiency; nowadays, there is a charm in digitisation. In some cases, this development leads to a change in what organisations produce, and how. ABB, Toyota, and Ericsson are increasingly becoming software companies, and their physical production is less and less manual. In other cases, the step from physical to digitised product and production logic is too large; those who were good at physical production, physical distribution, and physically performed services may find themselves being replaced by new organisations built on digitisation. The old postal services are not the leading actors in digital exchange. Camera and film companies, like Kodak, have been replaced by Samsung and Apple.

Digitisation presupposes choices of infrastructure. Half a century ago, it was important whether you chose IBM, Tandem, or Olivetti as your computer supplier because different suppliers' products were not compatible with each other; specific pieces of equipment would be suited—or less suited—for specific tasks, and the computer suppliers' future (and thus the future of the chosen infrastructure path) was uncertain. Today, the standardisation of hardware and networks has reached a level where such choices are no longer strategic to most organisations. Infrastructure is today normally a service rather than a good. Computation, storage, and transmission capacity are standardised services that can be bought from a host of global suppliers. The same goes for business applications: product databases, accounting, customer records, purchase history, webshops, etc., no longer need to be bought

as software; they are available as services that many suppliers can offer of similar quality, capacity and, in mature markets, at similar prices. As technical choices become less important, the important decisions tend to revolve around cost efficiency, availability, and security—and the ability to move your data to another service supplier if the contract terms or the functionality should be significantly better there (see Chapter 4 on different ways of outsourcing IT, and benefits and drawbacks associated with these).

Technical standardisation, and the supply of computer resources and infrastructure as standardised, easy-to-access services at competitive prices, is a trend that started some years ago. It makes the technical choices less decisive for what types of applications and what types of data one will be able to handle—and to modify as new needs arise. It also increases the visibility of how business-relevant IT decisions concern important business issues—which information one wishes to derive from one's information systems to support which activities, in order to achieve which goals. IT use is not an end in itself; at its core, it concerns which results or goals one wants to reach, and thus in which manner IT will be needed to achieve them. In the changeable world of today and tomorrow, we can rest assured that our needs, and perhaps even our goals, will change rapidly. These changes will partly stem from opportunities created by the technical development, but socially, economically, politically, ethically, and materially driven changes will also affect both business and goals. It will then be important to be prepared in such a way that the IT support needed continues to work and can be obtained in a manner that meets new or changing needs. Such preparedness will need to build on technical insights and skills, but above all, it will need to build on insights in, and monitoring of, the other sources of change.

The development towards standardised architectures and services also means that it becomes increasingly important to truly understand the business that is to be supported, even more so than to profoundly understand the technology delivering the support. We are not claiming that any insight into technological issues is now superfluous—it is difficult to choose wisely if the functions under consideration are completely black-boxed. But deep insight becomes less important in step with the increase in the number of suppliers who can deliver equivalent services. And business insight has always been important,

but the more the business builds on IT solutions and is intertwined with them (is digitised), the more important it becomes that the procurement of IT services builds on a solid understanding of the business needs, functioning, and resilience to disturbances.

The strategic IT issues used to be more of a computer-science nature. It was important to have methods for designing information systems, for maintaining them, and for further refining and developing them. Today, it is increasingly rather about developing the general strategic capability and decision-making in the organisation in order to achieve the organisational goals. IT becomes one of many support functions for achieving the goals, and IT-related decisions are being taken in more and more places in the organisation. Today, meaningful and valuable digitisation is everybody's responsibility.

1.2. IT's Informing Function

As noted above, IT can have different roles in an organisation. One aspect is how it contributes to realising the value proposition of the organisation, for example in relation to customers or suppliers, or as support for the co-workers' competence development. (This will be discussed further in Chapter 3.) Another aspect is how important IT is to the business: how mission-critical is it, and how much emphasis is there on being at the forefront of the use of IT? A coarse yet fundamental division of the roles of IT is into *rationalising* and *informing*.

IT's rationalising role—partly or entirely replacing manual labour in certain areas and thereby saving both time and money—has long been acknowledged. IT's role in generating, storing, and supplying information is also not new, as digital information systems have long been used to support the financial and operative reporting in organisations. But the last decades have brought a marked increase in the possibilities for data collection, analysis, and reporting, as well as a large set of tools and techniques, such as business intelligence, competitive intelligence, dashboards, data warehousing, data mining, and big data. This is a good reason for taking a closer look at how IT can contribute to informing. We will not place much emphasis on labels—they are broad, and there is rarely consensus on their exact meaning.

For example, business intelligence will be used to describe just about any kind of support for control and follow-up, with widely differing demands for functionality and much else that lies outside the transaction systems used at a given organisation. We are therefore going to concentrate on frequently occurring properties and functions of these types of systems, regardless of what the systems are called. What they have in common is that they provide data that should be possible to interpret into information about the business and its environment, so that co-workers at different levels can get a picture of how the business is working—and how it could work—and based on that make sound decisions and act in order to move the business in a desirable direction. To understand what is desirable, goals, standards, and other points of reference are needed. These can be both internal and external, historic and forward-looking, quantitative and qualitative. From this information perspective, some characteristics are particularly salient:

- Speed
- Integration
- Detail
- Orientation
- Accessibility

Speed. That information is provided reasonably quickly by means of IT could seem self-evident, but for a long time, there was a widespread view that information on managing organisations was not accessible sufficiently quickly. IT has certainly helped bring processing and communication times down. From the earlier process of monthly reporting being prepared manually by accountants, today, many people click themselves through to the latest figures on revenues and costs fairly soon after the events have taken place in the business. With the types of tools mentioned above, speed is even more pronounced: in many cases, there is even talk of real-time information provision. Production standstills and customers becoming dissatisfied are examples of events that it can be important to be informed of quickly. Automatic warnings can be sent to those responsible if a machine starts to malfunction or an important customer chooses to cancel a contract. This can enable responsive actions that might make a considerable difference. But speed

in information transfers also increases transparency. I can no longer handle my tasks or area of responsibility and choose what becomes visible to others. Many others within (and sometimes also outside) the business can also have become aware of the disruptions at the same time as me—perhaps even before.

What is considered as rapid depends on the type of information, who should receive it, and on the strategy and business models of the organisation. The production line that has come to a standstill or the scandal that has erupted can be important to know at once—minutes can count. To the stock trader, fractions of seconds can be important, since the transactions are carried out at the speed of lightning and business opportunities come and go at that pace. But for someone negotiating a large bridge construction contract, “quickly” probably equates to days.

Integration. If the information provided via IT was previously dispersed over several systems—for example, one for financial information, one for customer and market information, one for personnel records, etc.—we today find an increasing emphasis on the unified image that IT can help provide. This is not just about the practicalities of having access via an integrated system, rather than having to look up different pieces of information in different systems. More importantly, systems integration can help provide an image of connections between different parts of the business.

Financial results are practically always focussed, regardless of whether there is an ambition to maximise profits or if it suffices that revenues simply cover costs (see Chapter 3 on how the different parts of the business model can be related). But financial results can contain many different business events, and therefore do not provide complete signals regarding the health of the business. For this reason, it is increasingly common that results are measured in multiple ways within an organisation. Measurements and indicators along multiple dimensions can indicate connections between the different parts, and thereby teach the co-workers what drives, for example, profitability at different stages. Is there a clear connection between short processing times of cases and customer satisfaction, or between the number of sales meetings and the level of revenue? Is the combination of certain people on a project profitable? How does the distribution of staff across different roles match the needs that the commitments of the coming months will

pose? Information that is integrated from different parts of the business can also indicate clashes between different dimensions of the business. The successful decrease of lead times in production coincides with a dramatic increase in costs for certain purchased services. The emphasis on a new customer segment has given rise to helpdesk overload. The increased delivery precision does not at all show up in the customer satisfaction indicators. Such signals can also be important and provide bases for discussions about priorities.

Detail. For many purposes, financial information—such as balance sheets, income statements, and cash flow analyses—is too aggregated and abstract to provide sufficient insight regarding the business. Integrating it with other types of data can then be a way of providing a more concrete and nuanced picture. Also, the possibility to “drill down” (to look at constituent parts, possibly all the way down to individual transactions) that digitised data provide can help inform the user. Financial (and non-financial) data can often provide insights, but interpreting them is also likely to raise questions that require further investigation or at least an increased level of detail, for example regarding a particular employee, rather than an entire department; a well-defined customer group, rather than all customers; a particular day, rather than the entire month. It could be, starting from a cost item in the income statement, to click your way right down to the individual invoice that caused the variance. It could be to use more sophisticated cost allocation to generate insights regarding the profitability of different customer groups (perhaps one ought to direct more attention to some groups than to others?). Or it could be to visualise the customer-service call load hour-by-hour over the month so as to better plan the manning of the helpdesk.

Orientation. Data processing can also provide indications of what could happen, rather than just telling us what has already happened. A classic example of this is making suggestions to customers regarding other items they may want to buy, in addition to what they have already placed in their digital shopping cart. Another example is banks’ efforts to warn customers of possible frauds or to forecast which customers will probably not repay their loans (and through this forecasting, the ability to standardise the lending process more thoroughly). Retail chains can use weather forecasts to predict demand patterns for umbrellas, drinks, etc. Municipalities study migration and birth statistics to better

forecast the required dimensioning of day-care and schools. Other areas, too, have been impacted by forward-looking IT applications. The police in several countries use forecasting to determine whether criminal events are related and could have been committed by one and the same offender, and if—and when—new crimes may be expected. In healthcare, digitised models are used to predict the survival rates for different types of cancer.

Accessibility. If many are to be able to use “informing” IT tools as a normal part of their work, rather than these being confined to specialists, then intuitive presentation of information, requiring less user experience, is needed. It can also be important to present data in a visually appealing manner, for example using colours effectively to enhance the clarity of presentation. This may sound obvious, but at least previous generations of informing IT tools were often deficient in these respects. The development of fast storage and computation capacity now allows for both fast analyses of large amounts of data and the possibility to instantaneously click on to further analyses of what a figure builds on (as described in the “Detail” section above). Also, comparison presentations are becoming more common, and show relevant standards and historical values, thereby indicating variances and needs for action more clearly. What can be regarded as diverging and what action will be required to address the deviation, is of course not universally given, but will rather build on what has been agreed on in the organisation, and on subjective assessment. But standards can be built into systems and help generate automated suggestions for actions in specified situations.

The information characteristics presented above overlap to some extent. For example, integrated information can also lead to faster access and provide a more detailed picture. Depending on the extent to which the characteristics are emphasised, the informing role of IT can include anything from a clearer visualisation of existing data—for example a financial monthly report with coloured graphs—to forecasts of customer preferences, detection of competitor product launches, or support of decision processes. The affordances are not limited to structured data; unstructured data, such as free text and images, can also be included. Examples include searches in patent databases, product launches, etc., to become informed about what competitors are doing, to find potential

partners, or to identify talents to recruit. Another possibility is automated scanning of social media in order to assess how the organisation, or specific goods or services, are mentioned and discussed. This can lead to dialogues with customers and opinion leaders, to adjustments of the marketing, product design, complementing services, etc.

1.3. Information and Value

We want to emphasise that supplying information with the aid of modern IT does not automatically generate value. There are limitations in the systems and how they match the organisation. No tool solves all challenges and problems; flexible tools, too, have their areas of application, and their limitations. When investing in a system of this type, it is first crucial to determine the needs for structuration in the different decision situations in the organisation to ensure that adequate structure and correct bases are available to enable efficient and effective decision-making considering operational and business risks. Posing clear requirements on the decision processes and the bases for decisions enables the appropriate design of systems support for different types of decisions. Of course, however, it is difficult to determine exactly what information will be needed in the future.

We illustrate this with a case description of work to improve the accounting information setting at a construction company.

Construction Firm had a somewhat outdated accounting information setting, including a plethora of information systems that were deemed user-unfriendly. Accounting information here refers to a variety of financial and non-financial numbers that are compiled in an organisation to provide, for example, managers at different levels with an insight into how the business is progressing and bases for their decisions. It can encompass anything from the profitability of different products; staff cost in relation to total cost; the level of CO₂ emissions; the number of workplace accidents; and customer-satisfaction scores, to how many projects are finished on time—anything that people in an organisation choose to record to provide an image of the business. The same data existed in several places, and the choice of which

system to rely on depended on the habits in a given department, business area or geographical region. Sometimes, there was a need to access data from several systems in order to compose a more complete picture, for example in preparation for the yearly discussion of how to act on the market during the coming year. Which customers and products have the greatest potential and should be given the most attention? Financially: which profitability patterns can be discerned in different customer and product groups? Strategically: is there a reason to start constructing a number of environmentally certified houses to signal (or meet) environmental consciousness? Which events can be perceived on the horizon, for example, is any potential customer planning a large construction, and should we then submit a tender for it? What is the distribution of professions/positions? Are too many foremen approaching retirement, causing a risk of a lack of qualified personnel when large projects start in the coming years?

Since the data to explore such issues resided in different systems, it was manually extracted and compiled in Excel sheets. Even the quarterly reviews of the business demanded a great deal of manual work in order to achieve a clear view of economic status, customer relations, procurement, safety, etc.

The head office called for a “more professional information use”. Their ambition was, through new information systems, to both make the handling of management control information more efficient and create more relevant grounds for decisions. The latter would be accomplished by tailoring the information supply to managers based on level and department. Those managing production at a site probably need different information from those managing a district with hundreds of construction projects. By creating easily accessible accounting information views tailored to the specific management role, those at headquarters hoped that the managers would make better-informed decisions.

The head office initiators were keen to involve the members of the organisation in the work, so they performed solid investigations; surveys, interviews and workshops were used to capture how the managers experienced the accounting information setting and what their wishes for it were. Most of the views received concerned the need to simplify and make

the information setting more efficient; accounting information handling should be fast so time can be spent on other tasks. Views and search paths were felt to be complicated; sometimes time-consuming, non-intuitive search commands for reaching relevant data led to compact tables in black, grey and white. The managers expressed that they would rather have integrated systems, so that manual transfers of data would not be required. They also wanted to be able to quickly report using smartphones and hand terminals regardless of their location and to have appealing and logical user interfaces with, for example, illustrative warning flags.

Thus, there was no lack of opinions about the processing and presentation of data, but relatively few suggested types of information that they thought were missing. Some wanted a better overview of prices of purchased goods and services in order to compare districts, or more feedback regarding previous projects in order to create more realistic budgets for future projects, but many found it difficult to think afresh. To instead ask managers to describe specific decision instances in their work, and what information they felt they needed there, did not help; the majority still found it difficult to identify something that they lacked and that could help make them better informed. In addition, many line managers held the opinion that more information and analysis was not necessarily beneficial. Being a line manager largely means handling customer contacts, keeping a time-pressed production on track, and being present in operations, or handling unforeseen events. To sit in the office and ponder information competes with other tasks. It can even lead in the wrong direction. One type of project can be successful in one instance, and in the next work badly, depending on who from the construction company is involved, who the customer is, and which suppliers are contracted. To believe that a certain type of project is always profitable or that a supplier works well, just because some numbers from a previous project say so, was considered naïve.

As a first step in improving the accounting information setting, a Business Intelligence solution was introduced to collect and visualise financial and HR data. The tools did not

provide much new data; largely, existing data were provided. But they were presented in an appealing manner. For example, the financial results were presented in colourful graphs, while the HR tools collected dispersed information without missing any details. Even reminders of employees' birthdays were included. The BI tools were gradually rolled out and were appreciated. There was even talk of them in districts where access had not yet been provided, and even really sceptical managers had turned enthusiastic.

This case illustrates a number of necessary considerations in just about any organisation aiming to advance their use of the informing function of IT.

- It is typically easier to see the potential for a rationalising, rather than truly informing, use of IT. For example, making existing data and information timelier and more clearly visualised is more likely to come up as an improvement request than requests for retrieving entirely new types of data and information. The latter requires co-workers to think more outside the box.
- We humans cannot cope with unlimited amounts of information. There is a reason why we often term accounting information 'Key Performance Indicators'; the information should stress what is most important—key—in the business, not everything that is going on.
- Data and information are not sufficient *per se*. Rules, standards, and other means for sorting and evaluating data and information are also needed. For example, some tools can process data and summarise them in key indicators, but it is we who choose the indicators, and these choices matter; are we actually highlighting what is really key? Other tools sift through available data, summarising and selecting that which, according to set rules, is deemed worthy of attention and closer scrutiny. This could be sifting through a news flow to identify articles that deal with our organisation or with our competitors. It could be to find deviations from forecasts in structured data. The automated sifting decreases the need for

human toil and attention, but we still need to assess that which is being presented and act on it. What are the consequences for us of the expected product launch of a competitor in a year? Should we do something to benefit more greatly from the current praise of our sustainability drive? Is our employee turnover at an appropriate level? Is our profitability sufficient? Is it a problem that the number of customers in a specific segment is low?

In every organisation (and in its sub-operations and departments) there is a need to establish norms and to set standards for what is to be viewed as desirable and worth striving for, or undesirable, and what requires action, and what does not. A part of establishing norms and standards is to formulate strategies and/or business models (see Chapters 2 and 3). Thereby, one defines what is important to do and to achieve, and what is less so. It is thus not sufficient to compile information from different parts of the organisation and its environment; that information needs to be related to something. A systematic way of supporting the use of available information is through the design of structured decision processes. In Chapter 5, we will therefore take a closer look at how an integrated decision process can be structured to use all of the information and synthesise it in a systematic decision process. Finally, it is also important to consider what is important in the available information. How quickly should it be available? How detailed should it be? Who requires what? The needs differ between organisations, between roles, and between individuals, as demonstrated above.

1.4. Type of Business Determines IT Needs

All organisations are unique, to some extent. But there are of course also similarities and common denominators. All enterprises need some type of accounting. Everyone who has co-workers needs to keep track of who they are, just as all those who have recurring contacts with customers need to keep track of who these customers are. Those who have active contracts with customers or suppliers need to keep track of what these contracts contain. Just about every organisation needs ways of paying out and receiving money. And so on. When any such generic need becomes extensive, IT solutions can likely be useful.

But what hides behind these generic needs? In what ways do enterprises differ in manners that matter to the choice of IT support? Look at the following list of enterprise types. Does it seem likely that they require similar IT support?

- Service company, mass service (hotel, taxi, home care)
- Mass-production (assembly line, process industry, prefabricated housing)
- Customised (unique) production (design your own shoe at Nike, Sculpture, 3D printing of unique items, unique buildings [advanced bridge, palace, ...])
- Trading platforms (generic/special goods and services)
 - Generic goods and services, like Amazon, Alibaba; hotels.com, ticketing.com, Uber, Lyft
 - Special goods and services, like via eBay, AirBnB
- Payment solutions (Amazon Pay, PayPal, Standard Bank, Société Générale, Visa, MasterCard, Bitcoin)
- News agencies, media companies

Of course they require different kinds of support. The challenges for the hotel differ from those of a dairy plant, a marketplace for transportation or a news agency. For that reason, the strategic decisions concerning IT support made by each kind of business will also differ. And two similar enterprises may even face similar IT-support issues, but still ultimately make different decisions.

Another dimension is how much of the enterprise the IT-related decision should affect. Is it intended to support a specific department, a collaboration between departments, a rethinking of how to conduct the business (provided suitable IT support)? Or is it about transcending the own organisation's boundaries and supporting the interaction with customers and suppliers, or increasing the level of ambition even more, changing the roles of the collaborating partners? The unit that is facing automation of certain tasks; the product development group that is seeking better contact with and feedback from customers; the record company that moves from selling records to selling music files, or even to streaming the music via a subscription-based intermediary;

the consumer-goods company that starts to use crowdsourcing as a part of product development; the camera manufacturer that moves into organising and making the customers' pictures accessible: all these changes have, or can have, extensive IT elements that require strategic positioning. The considerations leading up to such decisions can be more of a technical nature, like scalability, operational reliability, and data security, or novel versus well-tried solutions. But to a large extent, they could be expected to be operations- and business-related, such as appropriate degree of automation; possible customer reactions and the own ability to process and deal with feedback from customers; the impact of alternative sales channels and price models on the profitability of the business; the build-up of competence to be able to attract and keep a community of constructively imaginative individuals; the ability to develop, launch and monetise appreciated picture-management services. This does not mean that the considerations are operations- and business-related *instead of* being IT-related; it is a matter of IT-related operations and business issues.

1.5. Digital Transformation and Fundamental Rethinking

Many high-profile, much talked about and highly-valued ventures that build on extensive use of IT have emerged since the turn of the millennium. Amazon, Alibaba, Spotify, Uber, and Lyft have reshaped business networks by acting as new middlemen—electronic platforms that enable those who want to offer products to reach potential customers and to do business with them. *Huffington Post*, Kickstarter, and Airbnb crowdsource from individuals to individuals to offer services that compete with established actors such as the BBC and *The Times*, BNP and Standard Bank, Hilton and Accor. All of these companies appear innovative and require both advanced IT solutions and positioning regarding business focus and competitive opportunities. It is no coincidence that the new business models are not presented by the existing large actors in these markets. They already have functioning, often profitable and large businesses that are worth maintaining, rather than radically changing. Rapid IT development can influence the organisational climate in IT-intense operations. But that does not

mean that all IT-related change is rapid. There can, as just noted, be commercial reasons for not rapidly and radically changing functioning operations and business models. Also, people are not machines, and it is the rule rather than the exception that compared with the rate of technical development, the rate of change involving a change of mindset and human behaviour is slow. Even if it is possible to build a new organisation with a business logic that diverges from that of existing competitors, it can still be highly challenging to convince intended customers and suppliers to accept the new approach as trustworthy, functional, and valuable.

When new models are launched and become successful, they are adapted by other actors who want to start similar ventures. Amazon's web bookshop soon elicited followers like the Swedish Bokus. Spotify inspired Apple to start Apple Music and the discount broker Charles Schwab showed the way for similar actors such as Avanza. For these followers/challengers, the strategic decisions to be made can differ somewhat from the challenges facing the innovators. Both the technical solutions and the business models already exist. The question is whether they can be improved and made more efficient, or if it is even possible for clones to enter the market and acquire customers without competing with the innovators. Sometimes, the IT solutions themselves can differentiate an actor, and be the reason why a customer (or a supplier) chooses one over another. Differentiation might lie in user friendliness or functionality. If so, then IT design is central. Alternatively, the IT solution may remain in the background, with the differentiation lying in the price model or range of the offer. In such cases, the IT design will only become relevant to competitiveness if it fails in some respect, for example if there are long or frequent service interruptions, incompatibility with new user devices, or insufficient security leading to the theft of user data.

IT is always used in a particular context. A department forms one part of an organisation, and has parallels in other enterprises. The organisation, in turn, is both a part of networks of customers, suppliers, and other stakeholders, and a part of a category of similar organisations—with or without competitive relations to each other. From a still larger perspective, we can see the organisation as a part of an ecology of enterprises, actors, institutions, and ideas which interact in ways that change over time. Today, there are not many ventures where smartphones

do not form a part of everyday life, and of the organisation's active or passive infrastructure. Crowdsourcing and servitisation are ideas that have arisen in different places and have come to spread increasingly in the business ecology (the wide and far-reaching business surrounding), not just close to where the ideas originated. Strategic decisions tend to be presented as unique, but normally, they are rather a case of being able to recognise a good idea and realise how it can be made to fit the circumstances in which one is operating.

1.6. Significance of Systematic Decision-making

In a complex environment, it is crucial to be able to make reasonable decisions in an organisation. Having larger or smaller sets of data will not suffice as bases for decisions, and will not help if systematic decision processes are lacking. This is true not least when it comes to formulating a strategy for reaching a set goal. The ability to make conscious decisions is thus absolutely vital. The central decisions that are made are often important for the business and associated with varying degrees of risk. It is also easy to give competitive advantages to other businesses that have a better overview of their decision-making processes. Decision situations are generally characterised by unique assumptions, poor structure, and high complexity. Despite this, rather few reflect on why they act in a certain manner. Instead, one guesses and seldom understands one's own motives. Nevertheless, one must make the decision to survive. So, what should the organisational decision process look like? When people in an organisation formulate a decision, they face numerous questions:

- What are the relevant alternatives? And how do we know that all important alternatives are included?
- What are the relevant consequences?
- Which views and people are important to pay heed to? How important are these people and their opinions? Which perspectives are important to consider?
- Which values are important in the decision setting? How can these values be assessed?
- How can we correctly estimate the probabilities?

- Do we need additional knowledge to make the necessary decisions?
- How do we know that a decision is well-founded and correct?

Based on such questions, we will keep returning to theories and practices about decisions and decision-making processes throughout this book. The reason for having a decision-making process is to achieve a better quality of decision. This is, however, not easy to assess, but a reasonable quality requirement is that the decision-making process provides more effective methods for collecting and analysing knowledge communicated to employees and other stakeholders and that it does not become too costly in relation to its value. You then must ask yourself how to judge whether the basis for a decision is solid enough, and how to estimate the value of additional information to supplement the basis. Regardless of whether the decision concerns a problem or an opportunity, it is important to know what you want to achieve. If the decision is well-founded and you follow up on its consequences, it will generate knowledge about the business and its environment that may be applied in future decision-making processes.

1.7. A Global Perspective

Digital transformation is an opportunity and enabler for companies, organisations, sectors, and institutions worldwide, not least in the developing world. It is not something you enter, implement, and conclude. It is not a destination; it is a permanent state of evolution that one enters and re-enters to achieve one's goals. From promising but expensive IT activities to be used for different purposes, understandings of the digital and its enabling components have matured together with several powerful technologies, where artificial intelligence (AI) is one component. Fuelled by fundamental needs in the developing world, digital transformation (and AI) has moved to the top of the agenda of key international organisations such as the United Nations, UNESCO, the European and the African Union.

Both the upside and the downside of digital transformation were forcefully demonstrated during the Covid-19 pandemic. For example, in higher education, digitisation has become a challenge, opportunity, and threat to which universities around the world must respond. In

value chains and value networks, digital transformation had allowed the development of logical coordination of geographically distant units spanning the globe, which proved vulnerable to physical and political disruptions during the pandemic and to political reactions to armed conflict and conflicting political ambitions. Thus, attempts at digital transformation should not be blind to physical and political aspects potentially affecting the transformation.

1.8. The Structure of the Book

We have noted that people in organisations act based on different time perspectives, both short-term and long-term. Acting for the long term is often viewed as acting strategically. The idea is that strategies should act as guidelines for the organisation in its short-term, daily operation. Digital transformation has potentially given us access to more information than ever to support the work, it enables automation of tasks, and can give rise to entirely new products, forms of enterprise, and ways of conducting business.

In the process of formulating and applying strategies, those responsible within the organisation need to include and reflect on how digitisation can contribute value to its long-term aims.

But what characterises organisational goals and strategy processes? And how are they affected by digitisation? We delve into these issues in Chapter 2. Then, in Chapter 3, we go on to illustrate how the concept of business models can contribute to the strategic discussion. Business modelling is about identifying activities that contribute to the realisation of strategies. We focus on how benefits can be achieved via digitisation, or even how digitisation can form the very foundation of a business model.

So far, the focus has been on strategies and business models, on goals and opportunities creation. In Chapter 4, we focus on how we can act to realise such opportunities. The basis is organising, how organisations can divide and coordinate tasks and decision-making for digitisation—within an organisation or between separate organisations. This involves roles to capture and realise the potential of digitisation, which role holders provide input and make decisions, competencies that are important for understanding the role of digitisation in the

business and how people's experience and competencies can be united in a fruitful, rather than conflict-laden, way in the digitisation effort. Next, in Chapters 5, 6, and 7, we return to the issue brought up above in Subchapter 1.6: how to achieve a better structure of grounds for decisions and of decisions themselves, how this can be applied in the important area of procurement, and how uncertainty and risk, which always accompany decision settings and planned activities, can be managed through probability and risk analyses.

Digitisation and strategy work often bring change to the organisation, and such change can be handled via projects. Therefore, the book deals with two aspects of projects. At a strategic level, managing the entire set of projects is important. Which projects should we run, which should we discontinue, and how does the portfolio of projects contribute to realising the organisation's strategy? Thus, the first project-related aspect is project portfolios from both practical and theoretical viewpoints (Chapter 8). Having provided this context, we then move on to the second aspect, the projects themselves: how projects can be conducted, what must be considered in the organisation regarding running projects, and methods and tools available for project managers (Chapter 9).

Chapter 10 focuses on the issue of globally sustainable digital transformation, specifically looking at universities. Universities face a major challenge if they are to contribute to the United Nations' Sustainable Development Goal 4, "to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all." In the eleventh and final chapter, we summarise the messages of the book. We start by exploring the business ecology perspective on digitisation that we have pointed to in previous chapters. Then, we relate what we have focused on in the book to an overarching model of organisational focus. From this wider scope, we return to our messages about strategies, goals and business models, decision-making and risks, and projects and project portfolios. To conclude, we remind readers of the catalyst metaphor with which we opened the book. Digitisation works as a catalyst—that which is good can become better, but that which is not functioning well and carefully designed can also effectively get worse. This is why it is so important to try to understand what the attempted digitisation should lead to, and how to work to keep guiding the process in the desired direction.